

Time travel

PHIL UA-78 Metaphysics

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A paradox about ability and causal loops

The story of Tim—a reminder about consistency

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You know, of course, roughly how the story of Tim must go on if it is to be consistent: he somehow fails. Since Tim didn't kill Grandfather in the "original" 1921, consistency demands that neither does he kill him in the "new" 1921. Why not? For some commonplace reason. Perhaps some noise distracts him at the last moment, perhaps he misses despite all his target practice, perhaps his nerve fails, perhaps he even feels a pang of unaccustomed mercy.

The puzzle: what abilities does Tim have?

Q: Is Tim, in 1921, *able* to kill Grandfather? One can make arguments on both sides.

- ▶ NB similar puzzles also arise in supposed cases of backwards causation without time travel.

An argument for 'yes': consider this other non-time-travelling character Tom, with the same intrinsic properties as Tim—same gun, skills, etc.—who planned to kill Grandfather's partner in 1921. Tom was able to kill the partner. But [controversial premise!] what a person is able to do depends only on the properties that Tim shares with Tom. So, if Tom is able to kill the partner, Tim is able to kill Grandfather.

Arguing that Tim is unable to kill Grandfather

People sometimes suggest that there is some general reason to deny ability that would arise in any story of time travel into the past, based on the principle that no-one is able to “change the past”.

This seems to be a confusion. Suppose Tim travelled back into the Jurassic Epoch, bringing a rifle; he was planning to shoot a dinosaur, but took pity on it at the last minute and didn't shoot. I don't see any interesting argument that Tim wasn't able to kill that dinosaur!

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So, we are looking for considerations specific to the Tim case.

Arguing that Tim is unable to kill Grandfather

One argument could be based on the idea that it is *metaphysically impossible* for Tim to kill Grandfather in 1921. If true, that would be a great reason to deny that Tim is able to do it—surely no-one can do anything metaphysically impossible!

But this seems dubious. Plausibly, there are metaphysically possible worlds like the following:

- ▶ A world where Grandfather is born 50 years earlier, fathers Tim senior in 1880, who then fathers Tim in 1900, who then kills Grandfather in 1921 (no time travel even needed!).
- ▶ A world where Grandfather is killed in 1921 but then miraculously later raised from the dead.

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When we are thinking about what it's metaphysically possible for Tim to do, the question of *origin essentialism* becomes quite important. Origin essentialists may think that it is metaphysically impossible for Tim to (concretely) exist without being

An argument from counterfactuals

You might be tempted to argue as follows:

- ▶ If Tim had killed Grandfather, Father would never have been born, and Tim would never have been born, so Tim would not have killed Grandfather. So if he had killed Grandfather, a contradiction would have been true. But people are not able to do things such that if they had done those things, contradictions would be true.

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Response to this argument: If P is metaphysically possible, 'If it had been that P, a contradiction would have been true' is false. To answer the question what would have happened if Tim had killed grandfather, we just have to figure out which of these worlds count as 'closest'.

(It is surely highly context-sensitive. . .)

A more promising argument from counterfactuals

Vihvelin's view: in the relevant sense of 'can', 'X can ϕ ' entails 'If X tried to ϕ , X might succeed'. But 'If Tim tried to kill Grandfather, he might succeed' is false. So Tim can't kill Grandfather.

- This premise is reminiscent of the classical counterfactual analysis of ability, according to which 'X can ϕ ' is *equivalent* to 'If X tried to ϕ , X would succeed in ϕ ing'. But Vihvelin only offers a necessary condition, and replaces 'would' with 'might'.

A worry about Vihvelin's premise: does 'She can clear her mind of all distractions' entail 'If she tried to clear her mind of all distractions, she might succeed'?

Lewis's view: 'can' is context-sensitive (or 'equivocal'); in some contexts 'Tim can kill Grandfather' is true; in others, it is false.

The puzzle of meeting one's former self

Two people or one?

A common trope: the time traveller goes back a relatively short time and “talks to their younger self” (or fights with them...).

An onlooker would say ‘There are two people having a conversation over there!’

An onlooker who said ‘A grey-haired guy just punched a dark-haired guy’ would surely be speaking the truth, while if they said ‘No, really a dark-haired guy punched a grey-haired guy’ they would be wrong.

Indeed, even the time traveller will be highly tempted to say things that seem to imply that there are two people involved in the conversation. ‘I know you don’t recognize me, but I know you very well’...

Two people or one?

There's an objection here to the claim that this was a successful case of time travel. For if it is, there is only one person involved, not two.

What is this person like, at the time in question? The usual B-theoretic idea is that times are 3D slices through the spacetime manifold, and the physical properties of a material object at a time are determined by the physical properties of the *intersection* of the object's 4D spacetime trajectory with that time.

If we want to say that there's only one person involved, this leads to the view that they are *spatially scattered*, weigh about twice as much as a regular person, are wearing 2 coats and 4 shoes. . .

Can context-sensitivity get us out of the problem?

Someone might suggest that while 'person' sometimes applies to objects with long spatiotemporal trajectories, it has a different use on which it applies to shorter-lived objects (of which there really are two involved in the conversation); and similarly for 'I', 'you', proper names, etc.

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Problem: 'I was once a baby' (and 'Every person involved in that conversation was once a baby') seem unambiguously true! But having once been a baby requires having baby-shaped parts of your trajectory.